

1987:

Dear Millson,

Nori brought me a copy of your paper with Goldman on deformations of flat bundles. It seems to me that it contains a number of vacuums, but also that, when connected, it gives the story results you ask for p 88-102-7. I begin with the latter.

The philosophy, which I had not realized before reading your paper, seems the following: in characteristic 0, a deformation problem is controlled by a differential graded Lie algebra ($d^2=0$), with quasi-isomorphic DG Lie algebras giving the same deformation theory. If the DG Lie algebra controlling a problem is "formal", i.e. quasi-isomorphic to $\oplus H^i$, then the versal (formal) deformation space is that of $\oplus H^i$, i.e. is the completion at 0 of the subscheme of H^1 defined by the equations $[u,v]=0$.

If one believes in this, what you do becomes transparent:

Deformation problem: G is a given reductive group, and one wants to

deform a given G -torsor P . Meant: $G(\mathbb{C})$ torsor, or, equivalently, $G(\mathbb{D})$ -torsor, with integrable connection ∇ .

Controlling DG algebra: (one is over $\mathbb{C}$):

$$\Omega^{**}(\mathrm{Lie} G^P)$$

If P can be reduced to a maximal compact $K \subset G$ (i.e. to a compact form of G), Hodge theory shows that the controlling

algebra is "formal" : one has quasi-isomorphisms of DG Lie algebras

$$(1) \quad \Omega^{**}(\text{Lie } G^P) \longleftrightarrow \text{Ker}(d') \longrightarrow \frac{\text{Ker}(d'')}{\text{Im}(d')}$$

and $\text{Ker}(d'')/\text{Im}(d')$ has $d=0$. To check the above, it is easiest to use the principle of the 2 types in the form that the double complex $\Omega^{**}(\text{Lie } G^P)$ is (forgetting $[\cdot, \cdot]$) a sum of

a) a double complex with $d' = d'' = 0$ ("harmonic part")

b) squares of isomorphisms:

$$\begin{array}{ccc} * & \xrightarrow{d''} & * \\ d' \uparrow & & \uparrow d'' \\ * & \xrightarrow{d'} & * \end{array} \quad (0 \text{ elsewhere})$$

As far as formal deformations are concerned, this concludes the story. It is not clear to me how to use Artin in the analytic case [sol. of anal. eqn.] to get the same for analytic-variety. The algebraic case is OK, but Artin is more powerful there.

Here is how to give a meaning to the philosophy (and check it)

Construction : How to attach to a DG Lie algebra L^* (over a char. 0 field k) a fibred category over the category of $\text{Sp}(A)$, $A \in \text{Mod } k$ - algebras of finite dimension with residue field k .

Object over $\text{Sp}(A)$: $[k = A/m]$

$$w \in L^0 \otimes m, \text{ such that } dw + \frac{1}{2} [w, w] = 0$$

Map : The Lie algebra L^0 give rise to a formal group $\hat{L}^0$; the A -points of this formal group are the $L^0 \otimes m$, viewed as a nilpotent Lie algebra, and equipped with Campbell-Hausdorff group law.

Notation : $\hat{L}^0(A)$.

The formal group $\widehat{\mathbb{G}}^0$ acts on L' (thanks to $[\cdot] : L' \otimes L' \rightarrow L'$), hence

$\widehat{\mathbb{G}}^0(A)$ on $L' \otimes A$. One has also a map

$$"g^{-1}dg" : \widehat{\mathbb{G}}^0 \rightarrow L^1 : \exp(\lambda) \mapsto \frac{1 - \exp(-\text{ad } \lambda)}{\text{ad } \lambda} [d\lambda]$$

$$u.e. : \widehat{\mathbb{G}}^0(A) \rightarrow L^1 \otimes m$$

One now defines a map from $w \in L^1 \otimes m$ to $w' \in L^1 \otimes m$ as being $g \in \widehat{\mathbb{G}}^0(A)$ such that

$$(2) \quad w' = g dg^{-1} + g(w)$$

One has to check that $dw' + \frac{1}{2}[w, w'] = g(dw + \frac{1}{2}[w, w])$ when w and w' are related by (1).

To be checked : for a square zero ideal I in A , and an object w on A/I , there is an obstructions in $H^2(L' \otimes I)$ to extend w to A ; if the obstruction vanish, isomorphism class of extensions form a principal homogeneous space under $H^1(L' \otimes I)$; automorphisms of any extension (trivial on A/I) are $H^0(L' \otimes I)$. This is functorial in L' , hence the fact that quasi-isomorphisms induce equivalences of fibred categories.

Application here Let X be an analytic variety and (V, ∇) be a (holomorphic) vector bundle with ∇ connection, $R=0$. The deformation problem is

$A \mapsto$ category of vector bundles with connection in the X direction, on $X \otimes \text{sp}(A)$, deforming (V, ∇) .

It is given by the previous construction applied to $\Omega^{**}(\text{End } V)$.

Req: In your application, because you have the explicit quasi-isomorphism (1), you have an explicit equivalence of fibred categories between

* deformations of (V, ∇) as above

* category with objects over $\mathrm{Sp}(A)$ the ^{pointed} maps

$$\mathrm{Sp}(A) \longrightarrow \text{subscheme of } H^1 L \text{ with } [u, u] = 0$$

and arrows: $g \in (H^1 L)^\wedge(A)$ from u to $g(u)$

The functor $\uparrow$ is such: a cohomology class $u \in H^1 L \otimes m$, with $u: \mathrm{Sp} A \rightarrow \text{scheme } H^1 L$, can be uniquely written as $w \in L \otimes m$ with $d'w = d''w = 0$. This w can be corrected by a $d'l^0$ to have zero curvature, if $[u, u] = 0$. The corrected $\tilde{w}$ is unique mod $\exp(\mathrm{Ker } d' \otimes m)$.

Req (1) is a filtered quasi-isomorphism, for the Hodge filtration. This should translate into saying that the pair of spaces

$$(\text{deformations of } (V, \nabla)) \supset (\text{deformations of } \nabla \text{ on a fixed } V)$$

is isomorphic to the intersection of $F^0 H^1$ with $F^1 H^2$ for the pair

$$\left(H^1 \supset F^0 H^1 \right) \cap \text{locus } [u, u] = 0.$$

II. Comments on your paper :

p1 l2 after co-B : no [take $\mathcal{G} = U_1$]

p6 l18 holomorphic? : I guess you want to start with η such that

$$d'\eta = d''\eta = 0$$

p7 l4 : why closed $\Rightarrow$ $(3,0)$ -component closed?

III. Generalisation

Hodge theory is available for variations of Hodge structure (complex variations, on compact Kähler manifolds). The useful bigrading of $\Omega^{*,*}(\text{Fid}V)$ is obtained by mixing that of Ω and that of $\text{Fid}V$. It is useful now that (1) is a filtered quasi-isomorphism. One gets

a) the same result as before on deformations of a representation of π_1 given by a complex variation

b) looking at the deformation theory controlled by $F^0(\dots)$: same result for the versal moduli of a filtered bundle with connection coming from a complex variation. One insists that the deformed connection must continue to satisfy "transversality" :
 $\nabla F^i \subset \Omega^1 \otimes F^{i-1}$.

Yours sincerely,

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